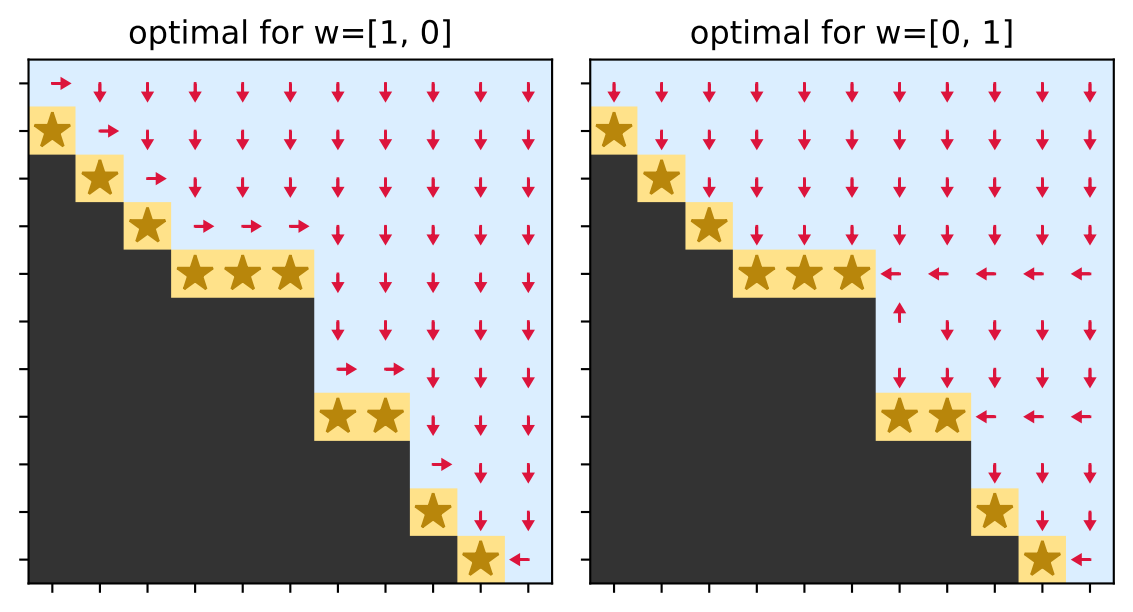
optimal for w=[1, 0]
optimal for w=[0, 1]